

Peak-Load Management in Commercial Systems With Electric Vehicles

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Abstract—Electric vehicles (EVs) are getting popular as one of the effective solutions for increased energy efficiency in commercial systems. This paper proposes an improved algorithm for commercial peak-load management using EVs, battery-energy-storage systems, and photovoltaic units. It uses the bidirectional vehicle-to-grid technique to utilize the energy from EVs in a parking lot. The proposed system has been tested in a real power distribution network in realistic load and weather conditions. The financial benefit of the system is also investigated, and it is found that the industrial peak load can be reduced by 50%, and the energy cost can be reduced by up to 27.3%. It also enhances the load factor by 9%. The performance of the proposed control algorithm is compared with that of an artificial-neural-network-based technique and tested in a laboratory prototype. From simulated and experimental results, it is found that the proposed approach provides substantial savings, while reducing the peak demand of the existing grids.

Index Terms—Commercial-load management, cost reduction, demand response, electric vehicle (EV), energy management, peak-load management.

NOMENCLATURE

c_{pv}^1	Capacity of a single PV module.
$c_{pv}^1 \cdot p_{pv}$	Supplied power from a single PV module.
x_p^{tp}	Electricity cost during peak hours.
x_p^{tb}	Electricity cost during OFF-peak hours.
y_p^{bat}	Battery energy cost.
y_p^{ev}	EV energy cost.
y_p^{pv}	PV energy cost.
$i_{b,e}$	Current from battery and EV.
i_p	Current from PV.
l_i^b	Power demand of the building at a particular time.
T_p^t	Duration of the peak-load period.
T_b^t	Duration of the OFF-peak (base) load period.
p_r^d	Required power to shave the peak at i th time.
p_a^c	Available power to charge battery and EV at i th time.

c_b^t	Battery SOC at i th time.
c_s^b	Maximum battery charging limit.
c_s^m	Maximum EV charging limit.
c_i^a	EV SOC at i th time ($n = 1, 2, \dots, N$).
β_{pv}^b	PV power provided to battery.
β_{pv}^g	PV power provided to grid.
COE	Cost to store the electricity (US\$/kWh).
AEP	Annual energy production by the storage (kWh/year).
BC	Battery-cell cost (annually).
PCSU	Unit cost of battery management and power electronics.
SUCU	Unit cost of the storage (US\$).
BOP	Total cost for balance of plant (US\$).
OMC	Annual fixed, maintenance, and installation cost.

I. INTRODUCTION

CONVENTIONAL power systems normally use bulky power generators and are designed to generate power by balancing the customers' demand. Usually, a customer's power demand varies over time based on their lifestyle and usage of appliances, resulting in fluctuations in the load curve. The minimum level of this load curve is known as the base load (OFF-peak) condition, where the maximum level is known as the peak-load condition. The frequency of this peak-load occurrence is low, and its duration is very small relative to the OFF-peak load. However, the power-system components, such as generators, transmission lines, transformers, and switchgear and protection systems, need to consider this small period and be designed with a higher capacity than their base-load capacity. Additionally, the high-ramp power generation for this small period costs much more than the average.

The present smart grid and the future grid, alternatively known as the Internet-of-Energy (IoE), will supersede the way power has been generated and managed in the conventional system. Both the smart grid and the IoE incorporate disaggregated and distributed renewable energy sources integrating intelligent controllers, prosumers, and new types of mobile loads. One such mobile load is an electric vehicle (EV), and in recent years, its penetration in the market is increasing rapidly.

Moreover, the battery capacity and the charging rate of EVs are increasing with a decreasing charging duration, resulting in a potential overloading of the power distribution networks and transformers. Other than this, mobile battery (EV), another type

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of battery, usually static, is increasingly used at the domestic point and at the grid side with renewable energy sources. Management of all these energy sources and loads has remained a big challenge for the present and future grid. However, if managed properly, these storages can provide various services, such as load management (OFF-peak charging and peak discharging), power quality [photovoltaic (PV) voltage rise and voltage flicker minimization], and power regulation (frequency and stability management, power flow balancing).

Battery-energy-storage systems are used in domestic systems and microgrids to get various ancillary services, such as frequency support, load support, and voltage support [1]–[5]. The effect of integrating large-scale battery-storage systems in the grid level is investigated in [6] and [7]. In some cases, EVs are used to get ancillary services [8]–[15]. Researchers have focused on PV and battery-based domestic energy-management systems [16]–[20]. At domestic levels, PV, EV, and battery are used to reduce the peak load using a rule-based algorithm [21]. None have investigated the peak-load management of commercial loads, such as shopping malls, hospitals, and schools, using a combined response from EV, PV, and battery storage. Additionally, the economic benefit of the load management has also not been analyzed in any level of load management.

In this paper, we consider this research gap, and use an integrated response from PV, EV, and battery to mitigate the peak of a commercial load. An improved algorithm is developed, and the proposed system is tested in a real power distribution network in Australia. In addition, a charge-scheduling algorithm is also developed. Real-time load and climate data are used to verify the algorithm. Furthermore, real-time load price data are used to investigate the financial benefit of the proposed system. The simulation results are also verified by a laboratory experiment.

In a nutshell, the contributions of this paper are as follows.

- 1) To design and implement an improved algorithm to minimize the commercial peak-load demand in a commercial system.
- 2) To develop an EV charge-scheduling algorithm.
- 3) To analyze the effect of coordinated response of the battery, EV, and PV in the peak-load management process.
- 4) To investigate the peak-load management system with real-time load, weather, and seasonal conditions in a real environment for a longer period.
- 5) To use an artificial-neural-network (ANN)-based technique to implement the proposed system and compare with the proposed algorithm.
- 6) To study the financial benefit of the peak-load management for a commercial load.

This paper is organized as follows. It begins with an overview of the proposed system with component description and specification in Section II. Section III describes the proposed EV management and load-management algorithm with a systematic flowchart. Various case studies are conducted in Section IV. Section V investigates the performance of the proposed algorithm in comparison with an existing one. Section VI provides a brief cost-benefit analysis of the proposed system with following experimental results in Section VII and conclusions in Section VIII.

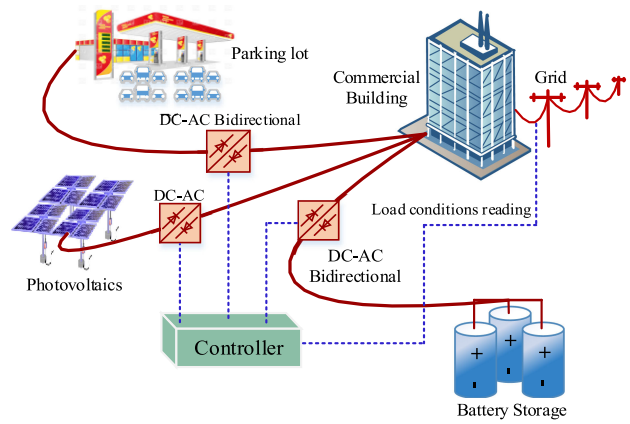


Fig. 1. Overview of the proposed system.

TABLE I
NUMERIC VALUES OF THE SYSTEM

Parameter	Value
Total number of EVs	6 (3 actively connected)
SOC of EVs to start V2G	from 85% to 95%
Current SOC of EVs	Random between 70–96%
Efficiency of EV chargers	96%

II. SYSTEM OVERVIEW

The system consists of six EVs in a parking lot, three PV cells, and a battery-storage system, as illustrated in Fig. 1. All these energy sources are connected to a common ac bus through power-electronic converters.

The controller reads the PV power generation, EV charge demand, EV state-of-charge (SOC), battery SOC, and building load conditions. Depending on the energy demand and the load, the controller controls the EV and battery charging and discharging cycles.

A. EVs in Parking Lot

It is assumed that the parking lot associated to the building has a parking capacity of 40 vehicles, and among them, 30% are EVs. Among the EVs, 50% are interested to participate in the vehicle-to-grid (V2G) operation, i.e., six EVs. It is assumed that among the EVs, 90% are Nissan leaf type, and 10% are Tesla type (in terms of capacity). The battery capacity of the Nissan leaf and Tesla EV is 24 and 90 kWh, respectively. It is assumed that the EVs at the parking lot stay 3 h on average in a random schedule from 7.30 am to 8 pm, and at any particular time, at least 50% of the EVs are connected to a V2G plug. All these assumptions are taken for the simulation analysis purpose only. General formulation of the EV's integration and load management is presented in Section III. Key parameters related to EVs are listed in Table I.

B. PVs and Battery

The proposed system assumes three PV cells on the rooftop of the building. The specifications of the PV and battery are listed in Table II.

TABLE II
NUMERIC VALUES OF PV AND BATTERY

Parameter	Value
No. of PV modules, each module capacity	3, 1.9 kW
PV capacity degradation per annum	1%
Efficiency of PV and battery converter	98%
Storage capacity	10-25 kWh
Battery lifecycle	10k
Battery charging-discharging efficiency	90%
Battery specific energy	60 MJ/kg
Battery self-discharge rate per day	0.2%

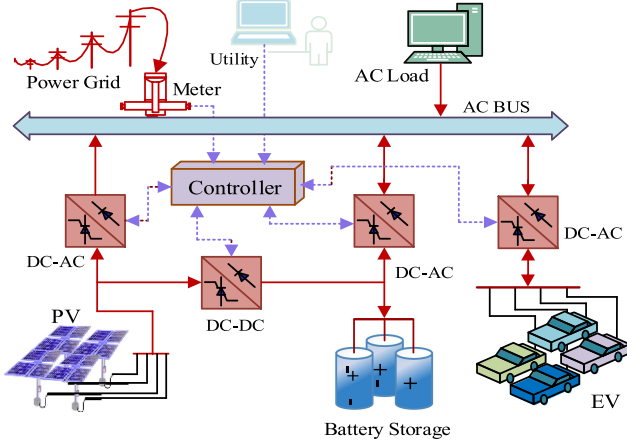


Fig. 2. Detailed architecture of the proposed system.

III. LOAD-MANAGEMENT ALGORITHM

In this section, algorithms for the individual components and the total system management are described. A detailed architecture of the proposed load-management system is shown in Fig. 2. All the energy sources (PV cells, EVs, and battery), the grid, and the loads are connected to a common ac bus. The PV is connected to the bus and battery through unidirectional dc-ac and dc-dc converters. The battery storage is connected to the bus via a bidirectional dc-ac converter. The EVs are connected to the bus through a V2G-enabled bidirectional dc-ac converter. The controller reads the PV power generation, battery SOC, and grid load conditions to control the power converters. It also reads the real-time electricity price from the utility to investigate the peak-shaved energy price.

A. EV Charge Management

The charging-discharging of the EVs connected to the parking lot is controlled based on the load conditions, SOC of the battery, the owner's set point (which means the desire of the owners to participate in bidirectional V2G energy transfer), and the charging-discharging limit. The charging-discharging limit can be set in the following two ways.

SOC based: EV owners can set the limit, i.e., up to which percentage of SOC, they want to charge the battery, or after which percentage of SOC, they want to sell (discharge) energy.

Travel-distance based: EV owners also can set the tentative travel distance, and the algorithm to charge and discharge the battery based on that trip plan.

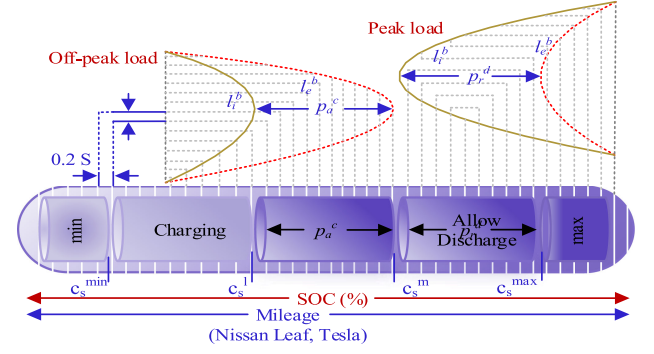


Fig. 3. EV battery charging-discharging process.

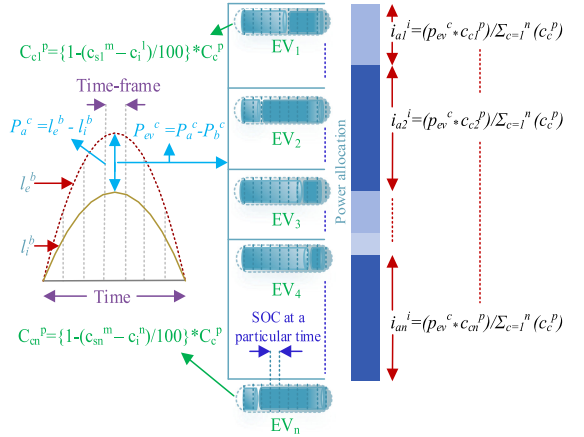


Fig. 4. EV charging priority and power allocation process.

Fig. 3 shows the available/required power for discharging/charging during peak and off-peak-load periods, with associated charge level constraints. A detailed schematic of the EV charge management is illustrated in Figs. 3 and 4, and the associated algorithm is described in Section III-B.

B. Total Load Management

The controller reads the load conditions and the battery SOC on a subsecond time scale, i.e., 0.2 s. However, due to a huge number of data in subsecond time scale (for 384 h) and to avoid the data overflow in meter, the meter reading at the commercial bus is read in 10-min interval. The battery and the EV charging and discharging are managed based on the load pattern, i.e., peak and off-peak-load conditions. The cycling of the battery is a function of the time set and power. Let the peak load occur between t_p^1 and t_p^2 , and the frequency of the occurrence be m , and the off-peak load occurs between t_b^1 and t_b^2 , with an occurrence frequency of n . If the power demand at a particular time is l_i^b , the grid load curve (T_l^t) can be expressed as the following functions:

$$T_p^t = m \{t_p^1, t_p^1 + 1, t_p^1 + 2, \dots, t_p^2\} \quad (1)$$

$$T_b^t = n \{t_b^1, t_b^1 + 1, t_b^1 + 2, \dots, t_b^2\} \quad (2)$$

$$T_{p,b}^t = (T_p^t \cup T_b^t) \quad (3)$$

$$T_l^t = \{T_{p,b}^t, l_i^b\} \quad (4)$$

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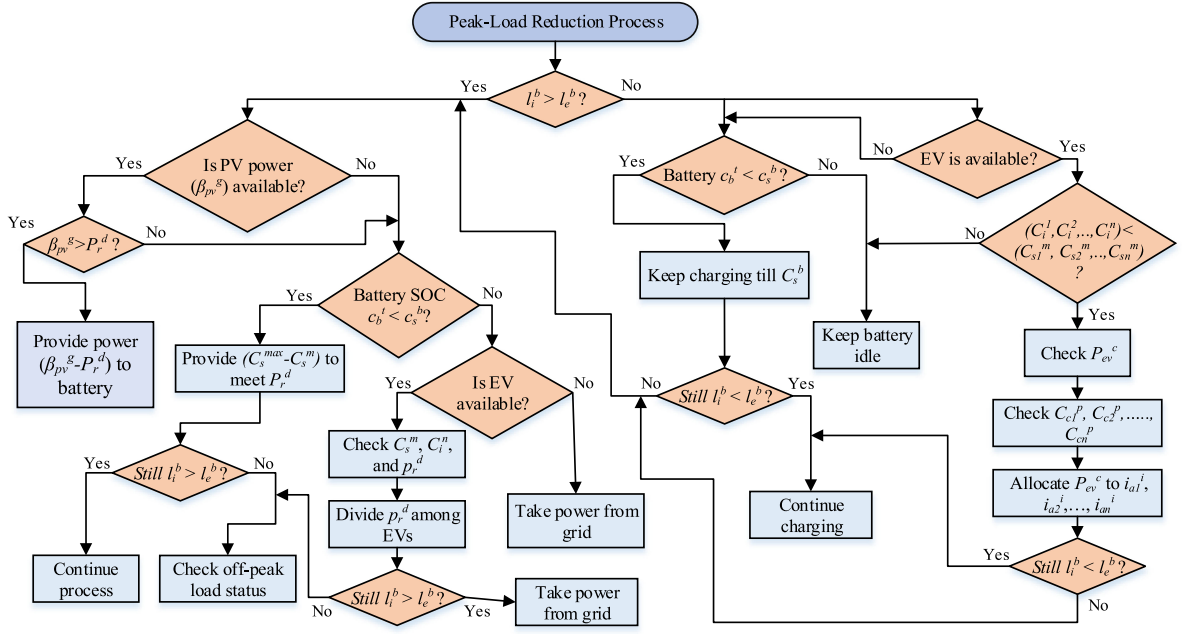


Fig. 5. Flowchart of the proposed peak-load management.

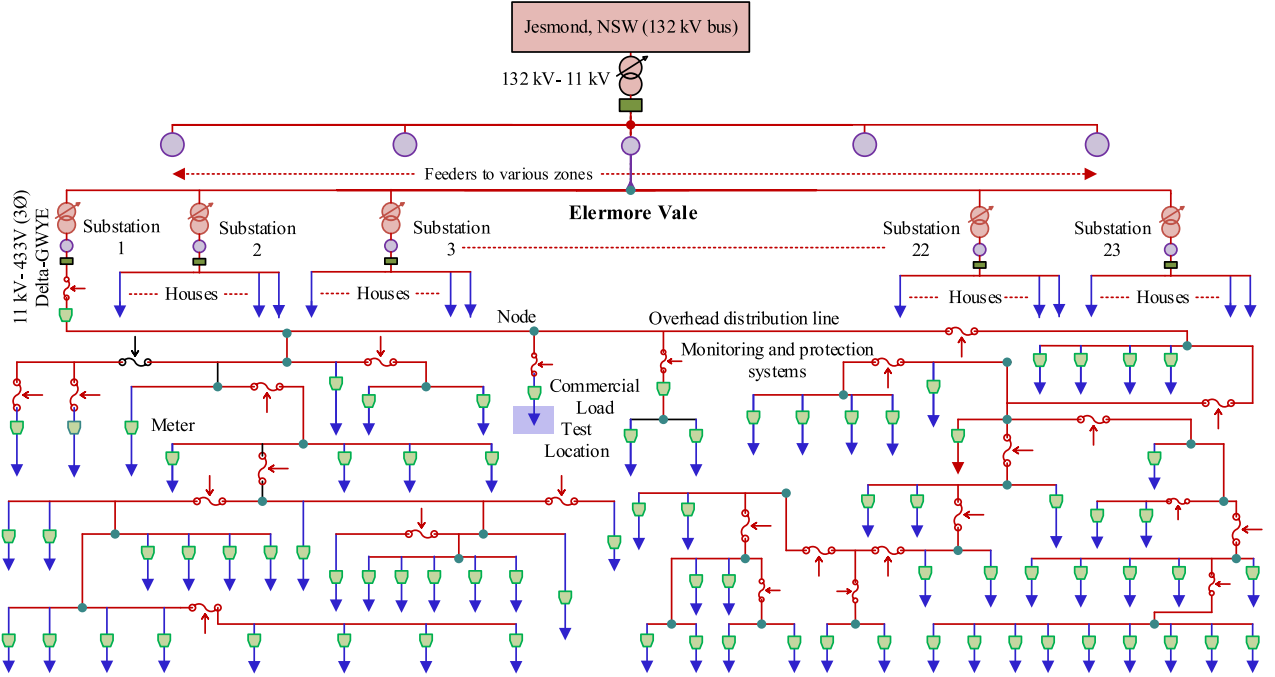


Fig. 6. Real network to test the proposed system.

TABLE III
AMOUNT OF PEAK-LOAD REDUCTION WITH VARIOUS COMBINATIONS
OF PE, EV, AND BATTERY

Combinations	Load factor (%)	Peak reduction (max %)
10 kWh battery only	43.88	11.80
10 kWh battery and PV	44.29	32.36
EV and PV	44.17	31.47
EV, PV and 20 kWh battery	47.29	45.56
EV, PV and 25 kWh battery	50.19	49.94
Without Proposed System	41.39	

following case studies is listed in Table III. The load factor and peak reduction are calculated as

$$\text{Load factor} = \frac{\text{Average load}}{\text{Max load at a particular time}} \quad (16)$$

Peak reduction

$$= \frac{(\text{Max load demand} - \text{load demand after control})}{(\text{Max load demand} - \text{average load demand})} \times 100\%. \quad (17)$$

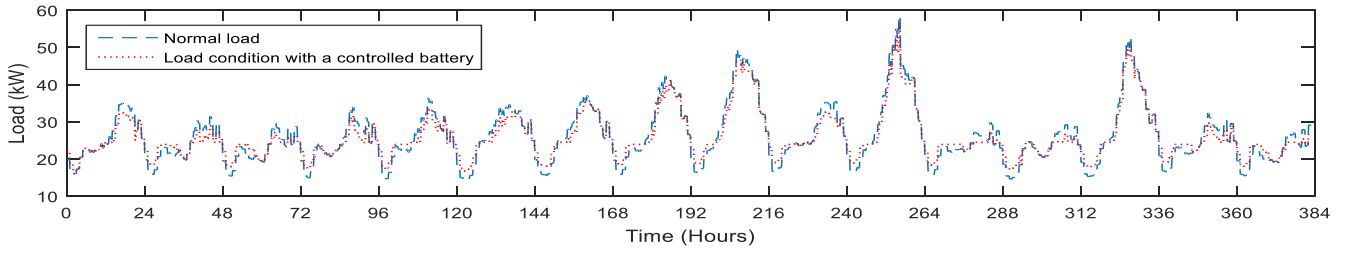


Fig. 7. Investigation of the peak-load reduction using a smaller size (10 kWh) of battery.

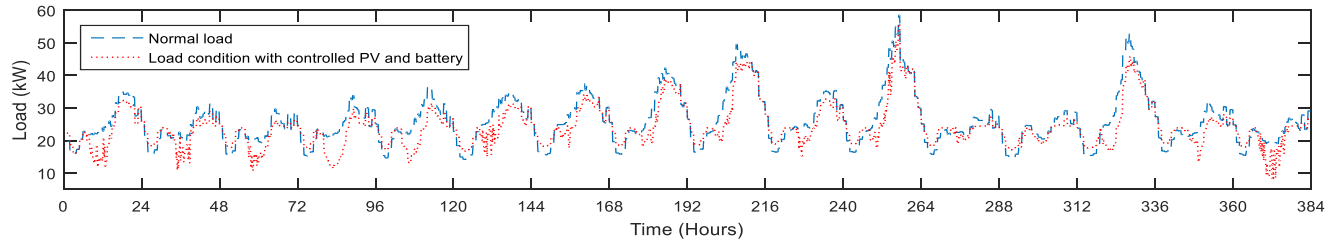


Fig. 8. Load curves with and without a smaller size (10 kWh) of battery and PV system.

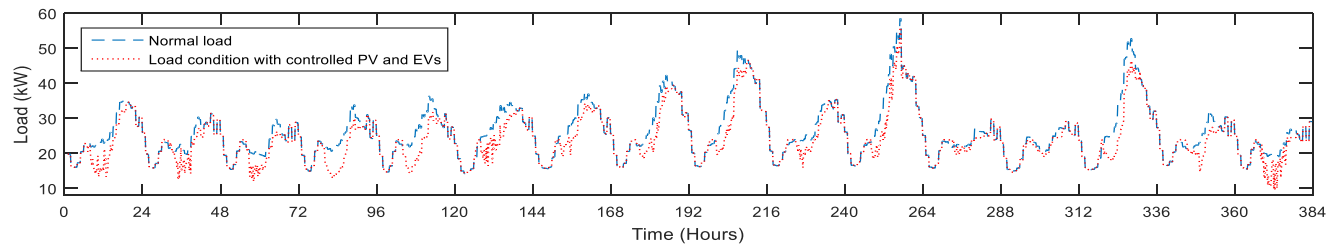


Fig. 9. Load curve with EVs and PV system.

The numerator in (17) (i.e., max load demand-load demand after control) represents the amounts of peak load that have been shaved by the proposed algorithm. The denominator represents the amount of peak demand that is expected to shave. Here, any load demand more than the average load (or base load) is identified as the peak load. In this paper, the base load is considered as 24 kW.

In this case study, only a smaller size of the battery (10 kWh) is used to minimize the peak of the commercial load. Initially, it is considered that PV is not available. The battery charging–discharging limit is kept between 45% SOC and 95% of SOC to enhance the lifetime of the battery. Due to the smaller capacity of the battery and the absence of the PV, the peak is only reduced up to 11.8%, as shown in Fig. 7.

When the battery storage is supported by a small capacity of PV, the peak load reduces more than the above amount. The parameters of the PV are listed in Table II, and the algorithm to share PV energy between grid and battery is described in Section III. From Fig. 8, it is clear that the proposed peak-load system works better with the battery attached to the PV system.

B. Peak-Load Reduction Using EVs

This section discusses the usage of EVs for peak-load management. The algorithm of the EVs integration is discussed in

Section III and the parameters are listed in Table I. From Fig. 9, it is clear that the EVs in a parking lot perform in a similar way to a battery. However, since the EVs stay in the parking lot from 7.30 am to 8 pm, they only participate in the peak-reduction process within this period. The charging energy needed within this period for some EVs is provided by the PV.

C. Peak-Load Reduction Using PV, EV, and Battery

In this case study, the combined response of the PV, EV, and battery is used to reduce the peak grid load of the commercial customers. A larger size of battery (20 and 25 kWh) is used, keeping the PV and EV capacity the same as before. Fig. 10 shows that the combined response of a 20-kWh battery, EV, and PV can reduce the peak load by up to 45.56%. The peak load is further reduced to 49.94% when battery capacity is increased to 25 kWh, as shown in Fig. 11.

D. Uncertainty Analysis

The uncertainty of the proposed system in the absence of EVs and renewables is tested under changing-load conditions. The loads used in the case studies (see Section IV-A–IV-C) are increased and decreased by 10%. The system performance due to 10% load increase is illustrated in Fig. 12, and the performance in 10% decrease is illustrated in Fig. 13. From the

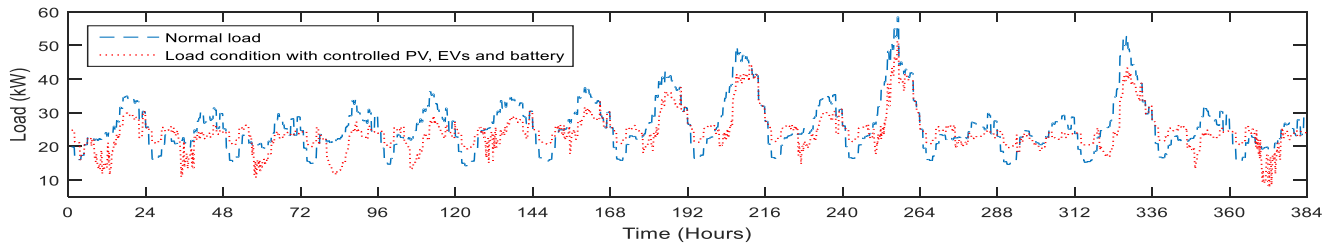


Fig. 10. Peak-load reduction by a combined response from 20-kWh battery, PV, and EVs.

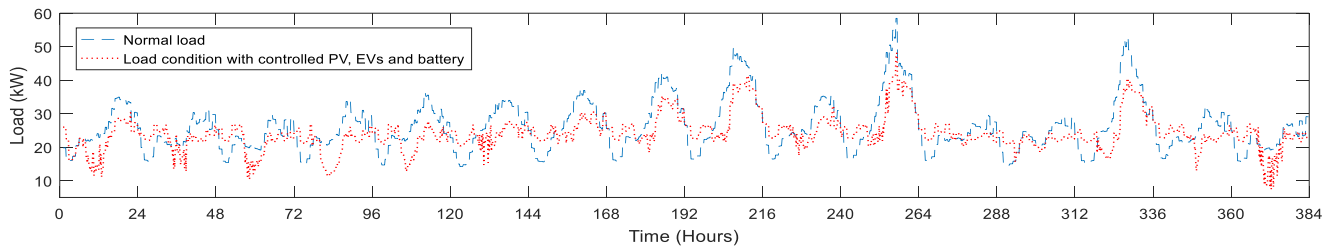


Fig. 11. Peak-load reduction by a combined response from 25-kWh battery, PV, and EVs.

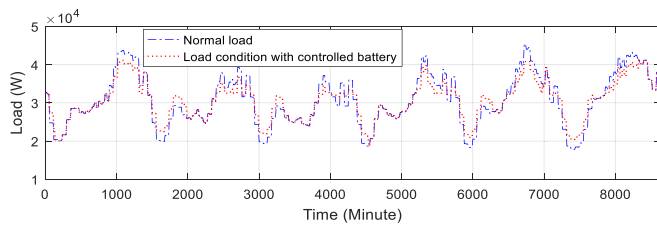


Fig. 12. Performance of the proposed system with a 15-kWh battery (EVs and PVs are absent) in a 10% load-increment condition.

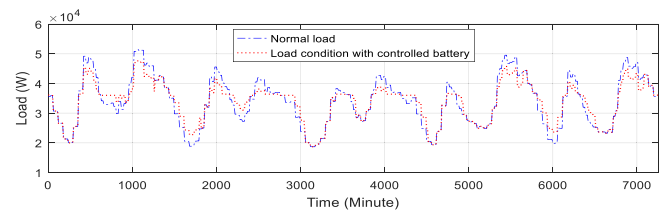


Fig. 14. Performance of the proposed system during summer season when only battery (15 kWh) is connected, and PVs and EVs are absent.

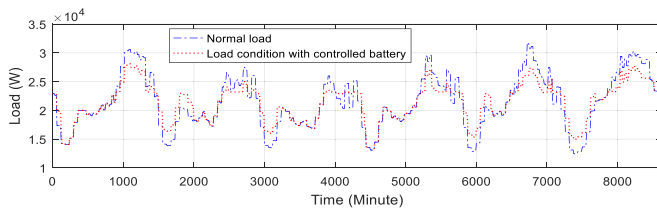


Fig. 13. Performance of the proposed system with a 15-kWh battery (EVs and PVs are absent) in a 10% load-decrement condition.

investigation, it is found that the proposed system is independent to load pattern and functions effectively based on the components' ratings and constraints. In case of the higher load conditions, small battery capacity will provide a small percentage of load shave based on the upper charging and lower discharging limit. Therefore, to achieve a high percentage of peak shave, a higher battery capacity is recommended.

The effectiveness of the proposed system is also tested under various load conditions in different seasons. The effectiveness of the system in the absence of EVs and PVs during the summer season is shown in Fig. 14.

V. COMPARISON OF THE PROPOSED SYSTEM WITH ANN

To validate the results, the performance of the proposed system is compared with that of an ANN-based system. An ANN can be used for both open-loop and closed-loop systems. Open loop is not adaptive and is often known as a feedforward neural network, whereas the closed-loop system can respond to parameter changes even after training the networks. The neural network technique known as a "nonlinear autoregressive with exogenous input (NARX) feedback network," implemented in [25] to estimate the capacity fade of the lithium-ion battery, is used in this system. 12 298 data have been used as the NARX target, and 62 298 real data were used as inputs for training. The input data consist of the load value, peak and off-peak-load conditions, battery SOC constraints, battery SOC conditions, charging-discharging conditions, and renewable energy-generation status. After the training process, various postprocessing analyses (mean-square error, error histogram, confusion matrix) were investigated to see the performance of the ANN. In this system, ten hidden layers have been used, and the system is trained after 91 iterations. The regression value during training and validation is 0.8913 and 0.884, respectively, where the system is most effective when this value is 1. The mean-square error during training, validation, and

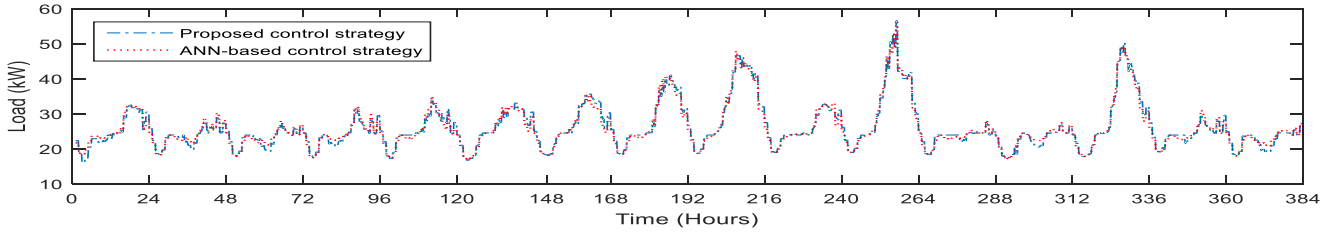


Fig. 15. Comparison of the proposed system with the ANN-based strategy.

testing is 2.4%, 2.3%, and 2.5%, respectively. The performance of the ANN-based controller using a 10-kWh battery to reduce the commercial peak grid load is shown in Fig. 15. The figure also compares the ANN-based peak-load performance with the proposed control strategy. It is clear from the figure that both the ANN-based strategy and the proposed control strategy give similar improved performance to reduce the peak grid load using the battery storage.

VI. COST-BENEFIT ANALYSIS

This section deals with a cost-benefit analysis of a commercial peak-load management system. The total cost to purchase and maintain a battery-storage system is divided by its net energy production to calculate the cost of electricity from energy storage. The total cost for storing per kilowatt of electricity can be expressed as

$$\text{COE} = \frac{(\text{BC} + \text{OMC}) - \text{SC}}{\text{AEP}}. \quad (18)$$

The total energy production per year (AEP) is the product of the capacity of the storage in rated power (in kilowatt), per day charge–discharge cycling number and their length (in hours), and the per year operating duration (in days), which is expressed as

$$\text{AEP} = P \times D \times \eta \times H_0. \quad (19)$$

The annualized capital cost for battery (BC) includes the per-unit battery-cell purchase cost, battery-management system and power converter cost, and the cost to balance the power. The total capital cost is expressed as

$$\text{BC} = \text{PCSU} + \text{SUCU} + \text{BOP}. \quad (20)$$

The total power-electronics converter, battery storage, and battery-management unit cost is the product of the unit cost with the battery-storage capacity and is given as

$$\text{PCSU} = \text{PCSU} \times P \quad (21)$$

$$\text{SUC} = \text{SUCU} \times P. \quad (22)$$

The operation, maintenance, and installation cost (OMC_f) per kWh of battery storage is multiplied by the total rated capacity of the storage system to find out the OMC which is defined as

$$\text{OMC} = \text{OMC}_f \times P. \quad (23)$$

The second life of battery storage is used in some applications. Therefore, the salvation cost (SC), which is the replaced battery storage cost, is deducted from the total energy-storage systems

cost in (18) and is provided as

$$\text{SC} = Q_0 - Q_{\text{loss}} \quad (24)$$

where Q_0 is the initial capacity of the battery and Q_{loss} is the capacity loss of the battery due to ageing, either by cycle fade or calendar fade.

Various battery operating factors, such as large depths of discharge (DOD), high and low temperatures, overcharging and discharging, high charging–discharging rates, and the formation of a solid-electrolyte interphase, are responsible for the battery ageing [27]. All these chemical and operating factors interact with each other, and, therefore, an appropriate battery ageing prediction becomes difficult. Various empirical and semi-empirical models have been proposed to predict the battery ageing [28]–[30].

In a semi-empirical model [31], a mathematical relationship between the capacity loss, temperature, charge throughput, and discharge C-rate of a lithium-ion phosphate cell has been proposed as

$$Q_{\text{loss}} = D \cdot \exp\left(\frac{-31700 + 370.3 \times C_{\text{rate}}}{G \cdot T}\right) (A_h)^{0.55} \quad (25)$$

where absolute temperature is expressed by T , gas constant by G , total charge throughput by A_h , capacity loss by Q_{loss} in percentage, and C-rate based preexamined factor by D .

Based on [32], the overall charging–discharging effect on battery life by considering the DoD and temperature is given as

$$L_{\text{ch-dis}} = \eta_{\text{DoD}} \cdot \eta_T \cdot L_N \quad (26)$$

where $L_{\text{ch-dis}}$ is the battery-life degradation, η_T is the temperature factor, η_{DoD} is the DoD factor, and L_N is the rated battery lifecycle at ambient temperature and rated DoD. The battery lifecycle reduction can be calculated by

$$\Delta L = L_N - L_{\text{ch-dis}}. \quad (27)$$

For a lithium-ion battery, the relationship between DoD and cycle life is logarithmic [32], and has the following relationships:

$$\ln(L) = -0.795 \ln(D) + 605425 \quad (28)$$

$$L = 694D^{-0.795} \quad (29)$$

where L denotes the number of life cycles of a battery and D denotes the DoD in percentage. Due to a lack of temperature data, in this research, only DoD is considered and temperature is considered ideal (i.e., ambient temperature).

The cost-benefit analysis illustrated in Fig. 16 considers the parameters listed in Table IV. The real-time electricity price data

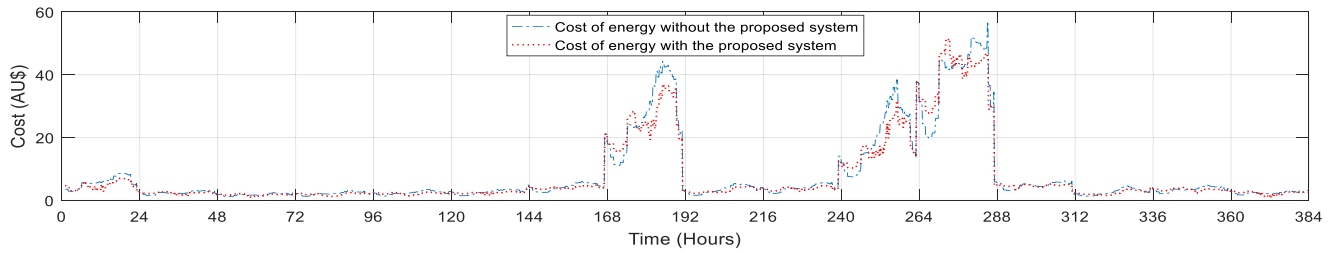


Fig. 16. Cost-benefit analysis of the proposed system.

TABLE IV
NUMERIC VALUES USED IN A COST-BENEFIT ANALYSIS

Parameter	Value
Battery cell cost	220- 240 AU\$ per kWh
Li-ion battery cycle life	>5000
Operation and maintenance cost	25 AU\$/kW-yr
Inverter, battery management cost	20 AU\$/kWh
PV panel installation cost	5-7k AU\$, for 3kW systems +5% maintenance cost, 15 years lifetime

provided by the Australian energy market operator (AEMO) have been fed to the controller of the system (as in Fig. 2) [33]. The electricity price is based on NSW, Australia, and is a time-of-use (ToU) energy price from January 30 to February 15, 2017. According to AEMO, the peak load and off-peak load have two different electricity prices. The peak load period is from 7 am to 10 pm and the off-peak load period is from 10 pm to 7 am, and weekends are considered as off-peak periods. AEMO fixes the electricity price to their customers based on their purchase from wholesale market for peak and off-peak periods of each day. From the graph, it is clear that the electricity cost can be reduced by up to 27.3% when the combination of PV, EV, and 25-kWh battery is used.

VII. EXPERIMENTAL RESULTS

An experiment has been carried out to investigate the proposed energy-management system. All the energy sources (batteries, PV, grid) are connected to a common ac bus as designed in the proposed system. The bus voltage is 230 V, like the Australian domestic supply voltage. A PV emulator is connected to the ac bus through an inverter. Four gel batteries (6 V each) are connected in series to make a total 24 V. This 24-V dc is then step up to 230-V ac to connect to the common bus through a battery inverter. A programmable ac load is connected to the bus to emulate the domestic power demand. Fig. 17(a) shows the total experimental setup. Since the bus voltage is constant (i.e., 230 V), the current varies based on the load conditions. In Fig. 17(b), the current required for the load is 5.09 A, which is supplied by the battery (3.58 A) and the PV (1.51 A). Alternatively, the power demand is 1.17 kW, which is supplied by the batteries (0.823 kW) and the PV (0.3473 kW). In another condition in Fig. 17(c), load demand is 5.15 A (1.185 kW) (including battery charging), which is supplied by the PV (1.51 A, 0.347 kW) and the grid (3.64 A, 0.837 kW).

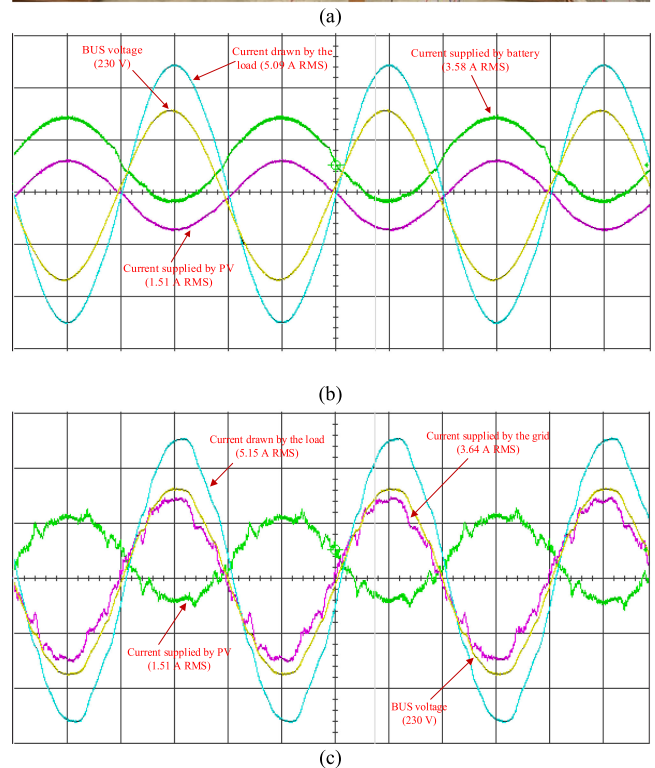
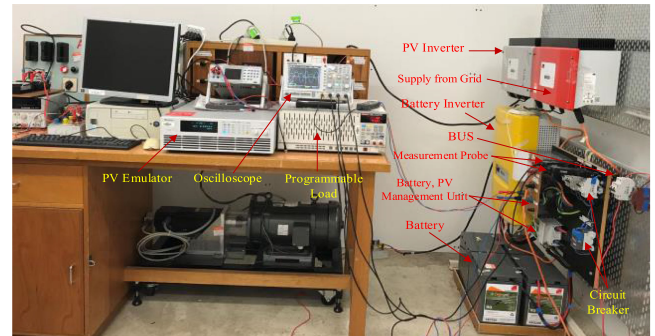


Fig. 17. (a) Experimental setup. (b) Peak load support by PV and battery. (c) Load support by PV and grid, and battery charging.

VIII. DISCUSSION AND CONCLUSION

The findings reported in this paper shed a new light on the utilization of EVs, battery storages, and renewables to reduce the electricity cost through the peak-load reduction process. The results of this study indicate that the proposed technique can significantly minimize the peak-grid demand of commercial

systems showing a robust behavior under various uncertainties in a real network and improve the load factor. Considering the battery degradation, both batteries and EVs use a narrow range of SOC and increasing this range will also increase the percentage of peak shave and decrease the required battery capacity. However, it will compromise the battery life. The battery-life-degradation calculation may vary slightly if all the factors associated with the battery life are considered. The cost-benefit analysis only considered the ToU tariff system due to the availability of its data. However, some other pricing systems, such as critical peak pricing and extreme day pricing, may provide more benefit.

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